# Javeria Azfar

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## Education

**Habib University** Aug 2021 - June 2025

*B.Sc. Computer Engineering, Minor in Social Development and Policy*
*HUTOPs Scholar (100% Scholarship)*

## Experience

**Habib University – Office of Research** Aug 2025 – Ongoing

*Researcher/Project Manager* *Karachi, Pakistan*

- Working on the perception, obstacle detection, and decision-making systems of an autonomous industrial tow truck running an Autoware / ROS 2 (Humble) stack in Docker, the first autonomous vehicle of its kind deployed in a Pakistani industrial setting.
- Contributed to a dual-Livox-MID360 3D-LiDAR perception layer that replaced a single forward-facing camera (87° horizontal FoV, no lateral or rear sensing) with full 360° coverage and no blind spots at 20 m+ detection range; field-validated for around-the-corner obstacle anticipation and full coverage through U-turns.
- Helped optimize sensor placement, inverting the LiDAR to bias vertical coverage toward the ground for improved detection of low obstacles (pallets, carts, curbs), and refined the perception codebase to reduce detection latency.
- Oversee the project's systems-engineering workflow in Innoslate (requirements, traceability, verification plans) and manage milestones, issue tracking, and sprint planning in Jira across the perception, control, and navigation sub-teams.
- Mentor and supervise three undergraduates developing components of the perception pipeline for their capstone projects.
- Collaborate with Toyota's engineering team on deployment testing and integration in a live industrial environment.

## Projects

**Developing a robust Vision System for Autonomous Tow Truck for Toyota Pakistan**

*ROS2, Gazebo, Python, Pytorch, C++* Fall 2024 - Spring 2025

- Developed and optimized real-time object detection and depth estimation pipelines using a customized YOLOv5 architecture and Intel RealSense D435i camera.
- Integrated visual perception modules into a **ROS 2 Nav2** framework for autonomous navigation and obstacle avoidance in semi-structured industrial environments.
- Containerized the full perception–planning system in **Docker** for reproducible deployment and modular testing.
- Conducted system validation through hardware-in-the-loop simulations in Gazebo and field testing at **IMC Port Qasim**.

**Diet Plan Optimization Using Evolutionary Algorithms** | *Python, MOGA* Spring 2024

- Built a MOGA generating optimized athlete diet plans balancing nutrition and performance; tuned fitness functions and evolutionary operators (elitism, crossover, mutation) for convergence and diversity.

**A Survey on Zero-Shot and Few-Shot Learning for Human-Motion Analysis using Vision-Language Models** |

Spring 2025

- Co-authored a survey of 30+ recent methods (e.g. MotionGPT, ZeDO, FG-MDM, PAML) across Human3.6M and AMASS using MPJPE, PA-MPJPE, and PCKh; proposed a taxonomy unifying meta-learning, generative, and optimization-based strategies.

## Publications

Azfar, I., Shahab, R., Azfar, J., & Raza, S.S. (2025). Optimizing Dietary Plans Using Evolutionary Algorithms. *Lecture Notes in Computer Science, vol 15612, (pp. 3-19).* https://doi.org/10.1007/978-3-031-90062-4_1

## Skills

**Programming Languages**: Python | C/C++ | Verilog | JavaScript | HTML/CSS | LaTeX| SQL
**Robotics and Perception**: ROS 2 (Humble) | Autoware | Gazebo | RViz | 3D LiDAR (Livox MID360) | NDT localization | point-cloud processing | RealSense RGB-D
**ML/CV**: PyTorch | TensorFlow | OpenCV | YOLO
**Tools and Workflow**: Linux | Docker | Git/GitHub | Jira | Innoslate (systems engineering)
**Domains/Research Interests**: Computer Vision | Robot Perception | Deep Learning | Reinforcement Learning | Human-Aware Navigation

## Awards

IFTP '23 2nd Runner-up - National, Kangroo Competition Bronze Medal winner - 2019 & 2020, AKU-EB Student Achievement Award - 2018